

# Irrationality after One Stabilization of Universally $CH_0$ -Trivial Cubic Threefolds

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## Abstract

Does universal  $CH_0$ -triviality force rationality after stabilization? We construct a non-isotrivial one-parameter family of smooth cubic threefolds  $X_b$  for which  $X_b$  is universally  $CH_0$ -trivial but  $X_b \times \mathbf{P}^1$  is irrational. More generally, we prove that  $X \times \mathbf{P}^1$  is irrational for every smooth cubic threefold  $X$ .

The cycle argument proves an all-degree integral saturation theorem for marked finite-etale graph quotients of elliptic powers: divided powers of the marked divisor lattice are ordinary divisor products. The six-axis intermediate Jacobians in the nonstandard  $A_5$ -invariant cubic pencil satisfy its hypotheses. Voisin's theorem gives universal  $CH_0$ -triviality on every smooth fibre; at non-CM fibres, the full integral Hodge lattice is divisor-generated.

For irrationality, we count primitive-sixth eigenvalues of framed formal monodromy on the numerical small even quantum connection. A divisor-tagging argument makes Iritani's blowup formula compatible with its possibly noninjective center specialization. The resulting count is a birational invariant through dimension four and equals two for a cubic. Kuznetsov's  $V_{14}$ -cubic flop then gives the same count and one-step irrationality for every smooth genus-eight prime Fano threefold. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

**Keywords.** cubic threefold; universal  $CH_0$ -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

## 1 Introduction

A smooth projective variety is universally  $CH_0$ -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension  $L/\mathbf{C}$ . This condition is a central consequence of stable rationality, and its failure is a standard stable-rationality obstruction. Its validity, however, need not provide a rational parametrization. We ask how far that gap can persist after one explicit stabilization.

Our examples are fourfolds of the form  $Y = X \times \mathbf{P}^1$ , where  $X$  is a smooth cubic threefold in the nonstandard  $A_5$ -invariant pencil. Two independent mechanisms act on the same  $Y$ . The intermediate Jacobian of  $X$  has an algebraic primitive minimal class; this gives universal  $CH_0$ -triviality. The small even quantum connection of every smooth cubic has primitive-sixth framed formal monodromy; Iritani's quantum blowup formula shows that its multiplicity survives one stabilization.

Here “nonstandard” has a precise representation-theoretic meaning. Let  $W_5$  be the irreducible five-dimensional complex representation of  $A_5$ , with character  $(5, 1, -1, 0, 0)$  on

the classes  $(1, 2A, 3A, 5A, 5B)$ . The symmetric-cube character formula gives

$$\dim(\mathrm{Sym}^3 W_5^*)^{A_5} = 2.$$

Thus  $\mathbf{P}((\mathrm{Sym}^3 W_5^*)^{A_5}) \cong \mathbf{P}^1$  is a pencil of cubic hypersurfaces in  $\mathbf{P}(W_5)$ . We denote its smooth locus by  $B^\circ$ . It is the complement of the discriminant in the pencil and is connected. This is Hartlieb's irreducible-character component  $M_{H_1}$ , as opposed to the component coming from the reducible permutation representation  $\mathbf{1} \oplus W_4$  [8, Lemma 5.5].

**Theorem 1.1** (Separation after one stabilization). *Let  $\mathcal{X} \rightarrow B^\circ$  be the smooth family in the nonstandard  $A_5$ -invariant pencil of cubic threefolds. For every  $b \in B^\circ(\mathbf{C})$ ,*

$$\mathcal{X}_b \text{ is universally } CH_0\text{-trivial,} \quad \mathcal{X}_b \times \mathbf{P}^1 \text{ is universally } CH_0\text{-trivial and irrational.}$$

*The family is non-isotrivial in coarse moduli.*

The irrationality statement does not depend on the symmetry of the pencil.

**Theorem 1.2** (One-step irrationality). *For every smooth complex cubic threefold  $X$ , the fourfold  $X \times \mathbf{P}^1$  is irrational.*

We do not claim that  $X$  is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

## How the two proofs meet

The cycle-theoretic half starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is  $6I - J$ , so any five axes give an isogeny  $E^5 \rightarrow J(X)$  of polarization type  $(1, 6, 6, 6, 6)$ . The two-primary principal kernels form  $\mathbf{P}^1(\mathbf{F}_4)$ . Three retain the full simplex symmetry; the other two are exchanged by Frobenius and have stabilizer  $A_5$ . The cycle argument needs only this unordered exotic pair. At two its graph slope has minimal polynomial  $t^2 + t + 1$ ; at three every  $A_5$ -stable graph is scalar. The graph equations turn both local divisor lattices into rank-one-generated matrix-of-ideals lattices. Because the corresponding rank-one divisors have square zero, every divided power expands as an ordinary product, including  $\Theta^4/4!$ .

The quantum half uses a different invariant. We count primitive sixth-root eigenvalues of framed formal monodromy on the numerical small even quantum connection. Cai's cubic calculation gives a nonzero count, and the projective-bundle formula doubles it on  $X \times \mathbf{P}^1$ . Iritani's blowup comparison is additive for this count. Its center Novikov map can be noninjective; formal divisor tags restore injectivity and the divisor equation returns to the original specialization. Surface classification then shows that every center of dimension at most two contributes zero.

Finite-etale graph saturation controls every divided power of the marked divisor lattice. For the actual six-axis packet, a separate four-slot calculation shows that the ambient integral Hodge lattice is also divisor-generated at non-CM fibres. The second argument is a direct weak-factorization calculation with the framed quantum invariant. The special  $A_5$ -geometry enters only in the first argument; the second applies to every smooth cubic threefold.

## Relation with earlier rationality results

Clemens and Griffiths proved that every smooth complex cubic threefold is irrational by means of its intermediate Jacobian [4]. That obstruction does not by itself control a product with projective space, and stable rationality of smooth cubic threefolds remains open. Cai recently recovered irrationality from the primitive-sixth-root formal monodromy of the cubic quantum connection [3]. Iritani and Iritani–Koto prove the quantum  $D$ -module decompositions for blowups and projective bundles [9, 11]; Katzarkov–Kontsevich–Pantev–Yu identify the nef-canonical normal form used for low-dimensional centers [12]. Theorem 1.2 combines these inputs at the next dimension.

On the cycle side, Voisin reduces universal  $CH_0$ -triviality of a cubic threefold to algebraicity of the primitive minimal class of its intermediate Jacobian [17]. Beckmann–de Gaay Fortman place such minimal classes in a general integral-Fourier framework [2]. For the very general cubic threefold the minimal class is not algebraic: every curve class on the intermediate Jacobian is an even multiple of it, so a very general cubic threefold admits no decomposition of the diagonal [6, Theorem 1.3 and Corollary 1.4]. What Voisin’s criterion leaves is the countable union of moduli subvarieties on which the class is algebraic; Theorem 3.5 places an explicit exceptional family inside that union. Here the six elliptic quotients supplied by Roulleau and the special period locus isolated by Hartlieb make the class explicit: here it belongs to the ordinary integral divisor-product lattice for every smooth member of the nonstandard  $A_5$ -pencil. The stronger equality of the integral Hodge and divisor-product lattices at non-CM fibres is proved by the same local graph calculation; the special CM fibres are not included in that stronger claim.

## Proof map

Section 2 constructs the rank-ten symplectic envelope and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.2 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

## 2 The integral envelope and the exotic cubic sheet

Let  $\Omega$  be the six conjugate  $D_5$ -subgroups of  $A_5$ , equivalently the six Sylow-5 subgroups with their normalizers. Put

$$A(\Omega) = \ker(\mathbf{Z}^\Omega \xrightarrow{\sum} \mathbf{Z}), \quad A(\Omega)^\vee = \operatorname{Hom}(A(\Omega), \mathbf{Z}).$$

The perfect evaluation pairing defines a unimodular alternating form on

$$\mathcal{H}_{\text{ref}}(\Omega) = A(\Omega)^\vee e \oplus A(\Omega)f, \quad \langle xe, yf \rangle = y(x).$$

This is the hyperbolic symplectic envelope of the oriented six-set.

**Proposition 2.1** (Universal envelope). *The construction  $\Omega \mapsto \mathcal{H}_{\text{ref}}(\Omega)$  is functorial for permutations of  $\Omega$ . Rationally it is the unique rank-ten symplectic  $A_5$ -module whose two Lagrangians are copies of the standard five-dimensional constituent and whose pairing restricts to the root-weight evaluation pairing.*

*Proof.* The permutation lattice and its augmentation kernel are functorial. The evaluation pairing between  $A(\Omega)$  and its dual is perfect, so the displayed alternating form is unimodular. Over  $\mathbf{Q}$ , the augmentation module is absolutely irreducible and its invariant symmetric

form is unique up to scale. Once the two specified Lagrangians are identified with the root constituent and its dual, Schur's lemma writes the rational symplectic module as  $A(\Omega)_{\mathbf{Q}} \otimes U$ , where  $U$  is a symplectic plane. The hypothesis that these retained integral Lagrangians pair perfectly then identifies them with  $A(\Omega)$  and  $A(\Omega)^\vee$ ; it is this hypothesis, not Schur's lemma alone, that fixes the root-weight lattice.  $\square$

The finite golden orientation enters at the prime two. Let

$$H_2 = \text{Aug}(\mathbf{F}_2^\Omega) / \langle \mathbf{1} \rangle.$$

Then  $H_2$  is simple and  $\text{End}_{A_5}(H_2) = \mathbf{F}_4$ . A normalized conference operator acts integrally on  $\mathbf{Z}^\Omega$ . After labeling  $\Omega$ , one switching representative is

$$B = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & -1 & -1 \\ 1 & 1 & 0 & -1 & 1 & -1 \\ 1 & 1 & -1 & 0 & -1 & 1 \\ 1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & -1 & 1 & 1 & 0 \end{pmatrix}, \quad B^2 = 5I.$$

Passing to the canonical saturation

$$D_6^\vee = \mathbf{Z}^\Omega + \mathbf{Z} \frac{\mathbf{1}}{2}, \quad \varphi = \frac{I+B}{2},$$

gives  $\varphi^2 - \varphi - 1 = 0$ . Its reduction on the commutator heart is one of the two primitive scalars  $\omega, \omega^2 \in \mathbf{F}_4$ , and replacing  $B$  by  $-B$  applies Frobenius.

**Proposition 2.2** (Golden selection of the exotic pair). *The two orientations of the conference class determine the two primitive scalars in  $\text{End}_{A_5}(H_2) = \mathbf{F}_4$ . Under orientation reversal,*

$$B \longmapsto -B, \quad \omega \longmapsto \omega^2.$$

*Thus the golden-orientation torsor is canonically the Frobenius torsor of the exotic two-primary principal gluings.*

*Proof.* Every column of  $(I+B)/2$  has the same half-integral class modulo  $\mathbf{Z}^\Omega$ , so  $D_6^\vee$  is the least stable over-lattice. The commutator submodule of  $D_6^\vee/2D_6^\vee$  is the four-dimensional six-point heart and its quotient is a trivial  $\mathbf{F}_4$ -line. The extension has no invariant  $\mathbf{F}_4$ -line, so the restriction of  $\bar{\varphi}$  to the heart is primitive. Finally  $(I-B)/2 = 1 - \varphi$ , and in characteristic two  $1 - \omega = \omega^2$ .  $\square$

We now recall the geometric source of the same lattice. Let  $S$  be the Fano surface of a smooth member of the  $A_5$ -pencil and  $(J, \Theta) = \text{Alb}(S)$  its intermediate Jacobian. Each of the six  $D_5$ -subgroups gives an elliptic quotient  $q_H : J \rightarrow E_H$  and its Rosati dual  $i_H : E_H \rightarrow J$ .

**Proposition 2.3** (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

*Any five axes therefore give an isogeny  $E^5 \rightarrow J$  whose pulled-back polarization has Smith type  $(1, 6, 6, 6, 6)$ .*

*Proof.* We recall the numerical passage from Roulleau's genus-two configuration to the six elliptic fibres. Let  $D_g \subset S$  be the curve indexed by an involution  $g \in A_5$ , put  $D_{\text{tot}} = \sum_g D_g$ , and, for a dihedral subgroup  $H$  of order ten, put

$$F_H = \sum_{g \in H, g^2=1} D_g.$$

Roulleau proves that  $F_H$  is a fibre of  $S \rightarrow E_H$  [16, Theorem 11(D)], so  $F_H^2 = 0$ . His intersection table has  $D_g^2 = -4$ , and distinct curves meet with multiplicity two or one when the product of their labels has order three or five, respectively. Among the 105 unordered pairs of involutions, the product orders two, three, and five occur 15, 30, 60 times. Hence

$$D_{\text{tot}}^2 = 15(-4) + 2(30 \cdot 2 + 60 \cdot 1) = 180.$$

Every involution belongs to two of the six dihedral subgroups, and therefore  $\sum_H F_H = 2D_{\text{tot}}$ . Two-transitivity on the six subgroups now gives

$$30(F_H \cdot F_{H'}) = (2D_{\text{tot}})^2 = 720, \quad F_H \cdot F_{H'} = 24 \quad (H \neq H').$$

After coherent identification of the generic non-CM elliptic quotients, two-transitivity makes the Rosati matrix have diagonal  $d$  and off-diagonal  $m$ . Put

$$D_H = q_H^*[0] \in \text{NS}(J), \quad N_H = i_H q_H \in \text{End}(J).$$

Then  $D_H|_S = F_H$ , and  $N_H$  is the Rosati endomorphism attached to  $D_H$ . Coherent identification gives  $q_H i_H = [d]$  and  $q_H i_{H'} = [m]$  for  $H \neq H'$ . The invariant sum of the six axis maps vanishes because the five-dimensional constituent has no trivial summand; composing with  $q_H$  gives  $d + 5m = 0$ . The minimal-class identity  $[S] = \Theta^3/3!$ , together with the polarized Riemann–Roch trace formula

$$\frac{\alpha\beta\Theta^{g-2}}{(g-2)!} = \text{tr}(N_\alpha) \text{tr}(N_\beta) - \text{tr}(N_\alpha N_\beta),$$

obtained by polarizing the reduced-norm Riemann–Roch polynomial [7, Theorem 4.1], gives

$$\text{tr}(N_H) = d, \quad \text{tr}(N_H N_{H'}) = m^2, \quad F_H F_{H'} = d^2 - m^2.$$

Thus  $d^2 - m^2 = 24$ . The positive solution is  $(d, m) = (5, -1)$ , hence  $G_6 = 6I - J$ . A five-by-five minor has Smith form  $(1, 6, 6, 6, 6)$ .  $\square$

**Proposition 2.4** (Principal gluing packet). *The  $A_5$ -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

*The three  $\mathbf{F}_2$ -rational kernels are  $S_6$ -stable. Each of the other two has stabilizer exactly  $A_5$ , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard  $A_5$ -cubic component uses the latter pair. Every  $A_5$ -stable three-primary kernel is a scalar graph.*

*Proof.* At two the coefficient discriminant is the simple four-dimensional  $\mathbf{F}_2 A_5$ -module  $H_2$ , whose commutant is  $\mathbf{F}_4$ . The full discriminant is  $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2$ . An  $A_5$ -stable simple half is therefore a one-dimensional  $\mathbf{F}_4$ -subspace of  $\mathbf{F}_4^2$ , and every such half is maximal isotropic for the tensor product of the coefficient form with the alternating multiplicity form. This gives  $\mathbf{P}^1(\mathbf{F}_4)$ .

Here is the polarization check explicitly. Identify the coefficient heart with  $\mathbf{F}_4^2$  and put

$$b(x, y) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2} \det(x, y).$$

On two multiplicity copies use

$$\Omega((x, y), (x', y')) = b(x, y') + b(y, x').$$

Every scalar  $a \in \mathbf{F}_4$  is self-adjoint for  $b$ , since  $b(ax, y) = b(x, ay)$ . Thus the vertical line and all four graphs  $\Gamma_a$  are isotropic of half dimension. The five module-theoretic possibilities are exactly the five polarized gluings; no further quadratic condition removes the exotic pair.

The simplex group acts on both multiplicity copies of the coefficient heart over  $\mathbf{F}_2$ . Hence the three lines defined over  $\mathbf{F}_2$  retain  $S_6$ . The stabilizer of an exotic graph is equally intrinsic. A permutation that fixes  $\Gamma_\omega$  commutes with  $\omega$ , hence is  $\mathbf{F}_4$ -linear on the two-dimensional  $\mathbf{F}_4$ -space  $H_2$ . If its determinant is  $d \in \mathbf{F}_4^\times$ , preservation of  $b$  says

$$\mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(dz) = \mathrm{Tr}_{\mathbf{F}_4/\mathbf{F}_2}(z) \quad (z \in \mathbf{F}_4).$$

The trace pairing is nondegenerate, so  $d = 1$ . It therefore lies in  $\mathrm{SL}_2(4)$ , of order 60. The given  $A_5$  already has that order and lies in the stabilizer, proving equality. The odd coset in its order-120 normalizer is semilinear and applies field Frobenius, so it exchanges  $\omega$  and  $\omega^2$ . Were the generic cubic kernel  $\mathbf{F}_2$ -rational, its principal polarization would inherit the faithful  $S_6$ -action. Strong Torelli identifies  $\mathrm{Aut}(J, \Theta)$  with  $\mathrm{Aut}(X) \times \langle -1 \rangle$ . Projection of this  $S_6$  to  $\mathrm{Aut}(X)$  remains faithful: its kernel would be a normal subgroup of  $S_6$  of order at most two, and no such nontrivial subgroup exists. This contradicts the generic automorphism group  $A_5$  on the nonstandard component [8, Section 5]. Thus the generic kernel is exotic. The three rational kernels and the two exotic kernels are distinguished by their stabilizers under the fibrewise  $A_5$ -action; they are therefore distinct monodromy-stable subsets of the five-element gluing packet. The geometric kernel is a finite sub-local-system, so generic exoticity and the connectedness of the smooth locus force it to remain in the exotic pair on every fibre.

At three the coefficient heart  $H_3$  is simple and  $\mathrm{End}_{A_5}(H_3) = \mathbf{F}_3$ . Hence its  $A_5$ -stable halves are the four graphs parametrized by  $\mathbf{P}^1(\mathbf{F}_3)$ , all scalar in a suitable graph chart.  $\square$

Proposition 2.2 labels the two exotic members by the finite golden orientation.

For the cycle-theoretic argument, the unordered pair is enough. Both members act on the heart with the same squarefree minimal polynomial  $t^2 + t + 1$ . The golden orientation supplies the finer marked identification with one member of the pair, but Theorem 3.5 uses only the conclusion that the geometric kernel is not one of the three  $\mathbf{F}_2$ -rational gluings. At three it uses only scalarity of the possible  $A_5$ -stable graphs, not the finer monodromy selection.

Hartlieb proves that its intermediate-Jacobian image is a one-dimensional special subvariety [8, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

### 3 Integral divisor products and the primitive minimal class

Let  $(A, \Theta)$  be a principally polarized abelian variety of dimension  $g$ . Put

$$P_A^k = \mathrm{im} \left( \mathrm{Sym}^k \mathrm{NS}(A) \longrightarrow H^{2k}(A, \mathbf{Z}) \right), \quad \gamma_A = \frac{\Theta^{g-1}}{(g-1)!}.$$

The class  $D^k/k!$  of a divisor on an abelian variety is integral. The question here is whether these divided powers already belong to the smaller lattice  $P_A^k$  generated by ordinary products of integral divisors.

For quotients of elliptic powers, this question is controlled by the valuation pattern of the descended coefficient forms. The main local result of this section says that finite-étale graph slopes force all divided powers of the prescribed graph divisor lattice into ordinary products, in every degree. The proof has two steps: the graph equations produce a symmetric matrix-of-ideals lattice, and a midpoint inequality makes that lattice rank-one generated. Rank-one coefficient divisors have square zero, so their divided powers expand without factorials. The same calculation will also show that the actual six-axis family has no ambient integral Hodge/product defect at its non-CM fibres.

### 3.1 A local graph theorem

Work over a complete unramified discrete valuation ring  $O$ , with uniformizer  $p$ . Let an elliptic-power source have coefficient polarization

$$G = PB, \quad P = \bigoplus_a p^a I_{M_a}, \quad B = \perp_a B_a, \quad (3.1)$$

where each  $B_a$  is unimodular. A *marked finite-étale graph presentation* consists of a ruling in which the principal kernel is the graph of an endomorphism  $T$ , with the following properties:  $T$  preserves the depth summands  $M_a$ ,

$$T^t B^{-1} = B^{-1} T \quad \text{or, equivalently,} \quad B T^t = T B, \quad (3.2)$$

and the algebra generated by  $T$  modulo  $p^a$  is finite étale on every positive-depth summand. Thus adjointness is taken for the dual coefficient form  $B^{-1}$ . No étaleness is required on  $M_0$ .

After a finite unramified extension splitting these étale algebras, the positive-depth summands decompose orthogonally into blocks

$$L_i = L_{a_i, \lambda_i}, \quad T_i = t_i I + p^{a_i} S_i, \quad (3.3)$$

with  $S_i$  integral. We keep the unit part as one block. Write  $\delta_{ij} = v_p(t_j - t_i)$ , with  $v_p(0) = \infty$ , for two positive-depth blocks.

**Lemma 3.1** (The graph coefficient lattice). *In the split basis, the coefficient lattice of descended graph divisors is the symmetric matrix-of-ideals lattice whose diagonal block  $i$  has depth  $p^{a_i}$  and whose  $(i, j)$ -block has depth*

$$e_{ij} = \max\{a_i, a_j, a_i + a_j - \delta_{ij}\}. \quad (3.4)$$

*For the unit block and a positive-depth block  $j$ , one has  $e_{0j} = a_j$ . In particular,*

$$e_{ij} \geq \max(a_i, a_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil. \quad (3.5)$$

*Proof.* In graph coordinates put

$$C = \begin{pmatrix} P^{-1} & P^{-1}T \\ 0 & I \end{pmatrix}.$$

For a symmetric source coefficient  $A$ , direct multiplication gives

$$C \begin{pmatrix} 0 & A \\ -A & 0 \end{pmatrix} C^t = \begin{pmatrix} P^{-1}(AT^t - TA)P^{-1} & P^{-1}A \\ -AP^{-1} & 0 \end{pmatrix}. \quad (3.6)$$

Thus descent is equivalent to the integrality of the three displayed nonzero blocks. Here  $A = A^t$  is a symmetric coefficient bilinear form in this normalization. On two positive-depth blocks  $L_i \times L_j$ , the two off-diagonal blocks first force  $A_{ij}$  to have depth  $\max(a_i, a_j)$ . Substituting  $T_i = t_i I + p^{a_i} S_i$  into the upper-left block leaves the additional condition

$$p^{-a_i - a_j} (t_j - t_i) A_{ij} \in M_{r_i \times r_j}(O);$$

the terms containing  $S_i, S_j$  are already integral. Intersecting the two conditions gives (3.4). Changing a scalar lift cannot change its maximum. Pairing the unsplit unit block with  $L_j$  gives  $e_{0j} = a_j$  directly from (3.6), and (3.5) follows.  $\square$

The inequality (3.5) is the valuation shadow of the tropical positive semidefinite cone studied by Yu [18]. We need the following signed integral lift of that inequality; in particular, no division by two is permitted.

**Lemma 3.2** (Rank-one generation over a DVR). *Let  $L$  be a symmetric matrix-of-ideals lattice over  $O$ , with diagonal depths  $p^{a_i}$  and cross depths  $p^{e_{ij}}$ . Then  $L$  is generated by its rank-one forms if and only if*

$$2e_{ij} \geq a_i + a_j \quad (i \neq j). \quad (3.7)$$

*Proof.* If  $cvv^t \in L$ , the two diagonal conditions imply

$$v_p(cv_i v_j) \geq \left\lceil \frac{a_i + a_j}{2} \right\rceil.$$

Hence rank-one forms cannot generate a shallower cross ideal.

Conversely, fix a generator  $p^e(uv^t + vu^t)$  of a cross block. Choose  $x \geq a_i$  with  $y = 2e - x \geq a_j$ . The integers  $x, y$  have the same parity. If  $t = \min(x, y)$ , write  $x = t + 2r$  and  $y = t + 2s$ . Then

$$p^t(p^r u + p^s v)(p^r u + p^s v)^t - p^x u u^t - p^y v v^t = p^e(uv^t + vu^t). \quad (3.8)$$

All three rank-one terms lie in  $L$ . Together with the diagonal rank-one forms, these identities generate  $L$ , without division by two.  $\square$

**Theorem 3.3** (All-degree graph saturation). *For a marked finite-etale graph presentation, every divided power of a class in its prescribed graph divisor lattice is an ordinary integral divisor product. Equivalently, after localization at  $p$ , the divided-power envelope of that lattice and its ordinary product image agree in every degree.*

*Proof.* After the unramified splitting extension, Lemmas 3.1 and 3.2 express every class in the extended graph divisor lattice as a finite sum  $D = R_1 + \cdots + R_m$  of rank-one coefficient classes. These are classes in the coefficient extension of the original lattice, not newly descended line bundles. On the elliptic-power source the alternating form of each  $R_i$  has rank two, hence  $R_i^2 = 0$ . Pullback by the isogeny is injective on torsion-free cohomology, so the same identity holds on the quotient. Therefore

$$\frac{D^k}{k!} = \sum_{1 \leq i_1 < \cdots < i_k \leq m} R_{i_1} \cdots R_{i_k}, \quad (3.9)$$

an  $O'$ -linear combination of ordinary products over the splitting ring  $O'$ . More precisely,

$$P_{\text{gr}}^k \otimes_O O' = \text{im} \left( \text{Sym}^k(N_{\text{gr}} \otimes_O O') \longrightarrow H^{2k} \otimes_O O' \right).$$

Faithful flatness descends the quotient of these finite lattices, and hence descends (3.9) to  $\mathbf{Z}_p$ .  $\square$



The theorem concerns the divisor lattice selected by the marked elliptic-power presentation. If  $E$  has no complex multiplication, this is the full Neron–Severi lattice. At a CM fibre extra divisor classes need not satisfy the graph hypotheses. The statement is cohomological: it neither constructs Chow divided powers nor asserts that every integral Hodge class is a divisor product.

For comparison with the classical rational theory, Milne proves that the Lefschetz classes are the rational algebra generated by divisors [14, Theorem 3.2]. Divided powers in Chow and integral Fourier transforms are studied in [15, 2]. The local result above instead keeps the weighted integral graph lattice and the ordinary, rather than divided, product image.

### 3.2 The six-axis packet

We now apply the graph theorem to the polarization computed in Section 2.

**Lemma 3.4** (Local chart for the six-axis quotient). *Let  $G = 6I_5 - J_5$ . At each prime  $p \mid 6$ , the coefficient lattice has an orthogonal decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where  $U_0$  has rank one and  $U_1$  rank four, both unimodular. The principal quotient is trivial on  $U_0$ . On  $U_1/pU_1$ , its graph slope has minimal polynomial  $t^2 + t + 1$  at  $p = 2$ ; at  $p = 3$  it is a scalar, hence a one-point finite-étale block.

*Proof.* Let  $e_1, \dots, e_5$  be the standard basis. Since  $\langle e_1, e_1 \rangle = 5$  and  $\langle e_1, e_j \rangle = -1$ , the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of  $e_1$  over  $\mathbf{Z}_p$ . Its Gram matrix is  $\frac{6}{5}(5I_4 - J_4)$ , and  $\det(5I_4 - J_4) = 5^3$ . This gives the asserted unit line and depth-one block at two and three.

At two, Proposition 2.2 identifies the selected action on the discriminant heart with  $\omega \in \mathbf{F}_4$ , whose polynomial is  $t^2 + t + 1$ . At three the coefficient heart is simple with scalar commutant. These are the actual kernels of the six-axis quotient, so the local presentations persist throughout each connected smooth component.  $\square$

**Theorem 3.5** (Divided powers for the six-axis family). *For every smooth member  $X_b$  of the nonstandard  $A_5$ -cubic pencil, the prescribed six-axis graph divisor lattice on  $J_b$  has all its integral divided powers in the ordinary divisor-product image. In particular,*

$$\frac{\Theta_b^4}{4!} \in P_{J_b}^4. \quad (3.10)$$

*Proof.* Only the primes two and three occur in the Smith form  $(1, 6, 6, 6, 6)$ . Lemma 3.4 verifies the hypotheses of Theorem 3.3 at both primes. At every other prime the isogeny identifies the integral source and target lattices. Since the relevant cokernels are finitely generated, the local memberships give the integral assertion. The polarization itself belongs to the prescribed graph lattice, yielding (3.9). Extra Neron–Severi classes at a special fibre can only enlarge the ordinary product image needed for this last assertion.  $\square$

At non-CM fibres one can determine the entire ambient integral Hodge lattice, not only the divided-power envelope. Write

$$I_A^k = H^{2k}(A, \mathbf{Z}) \cap H^{k,k}(A), \quad Q_A^k = I_A^k / P_A^k.$$

**Lemma 3.6** (Integral multidegrees on an elliptic power). *Let  $E$  be a non-CM elliptic curve. On  $E^g$ , every integral Hodge multidegree in codimension  $k$  has type*

$$(2^{k-\ell}, 1^{2\ell}).$$

*Its invariant lattice is a primitive product of  $k-\ell$  diagonal volume classes with the matching lattice on the remaining  $2\ell$  slots. The noncrossing matchings form an integral basis, with the three-term Plucker relations as integral straightening relations.*

*Proof.* The Hodge group is the diagonal  $\mathrm{SL}_2$  on  $H^1(E, \mathbf{Z})^{\oplus g}$ . The integral first fundamental theorem expresses its invariants as contractions. Sorting a contraction by coefficient-slot degree produces the displayed multidegrees: a doubled slot contributes its primitive volume class, and the single slots are paired. The standard noncrossing basis and unit-coefficient straightening are the two-row case of the integral Hodge-algebra basis [5, Theorem 11.1].  $\square$

**Theorem 3.7** (Integral Hodge saturation of the six-axis packet). *If the common elliptic factor in the six-axis presentation of  $J_b$  has no complex multiplication, then*

$$I_{J_b}^k = P_{J_b}^k \quad (0 \leq k \leq 5). \quad (3.11)$$

*Proof.* Lemma 3.6 shows that no prime away from the isogeny degree contributes to  $Q_{J_b}^k$ .

It remains to inspect two and three. In split graph coordinates, the off-diagonal divisor on slots  $i, j$  is, up to a unit,

$$\Omega_{ij} = p^{-a_i - a_j} (t_j - t_i) X_i X_j + p^{-a_i} X_i Y_j + p^{-a_j} X_j Y_i. \quad (3.12)$$

Here  $(X_i, Y_i)$  is a symplectic cohomology basis on slot  $i$ ; the integral off-diagonal divisor is  $D_{ij} = p^{e_{ij}} \Omega_{ij}$ . Only a squarefree four-slot component can fail to be generated by products. Its invariant space has the two matching directions, related by the three-term Plucker identity. The possible supports and their product scales are

| $p$ | four-slot support            | scales of the two matching directions |
|-----|------------------------------|---------------------------------------|
| 2   | $A, A, B, B$                 | $p^2, p^4$                            |
| 2   | $0, A, A, B$ or $0, A, B, B$ | $p^2, p^3$                            |
| 3   | $A, A, A, A$ or $0, A, A, A$ | $p^2, p^2$                            |

(3.13)

Here 0 denotes the unit line with its trivial graph coordinate  $t_0 = 0$ , and at two the depth-one roots occur with multiplicities  $2 + 2$ . In the two dyadic rows a repeated-root matching realizes the minimal scale and has no all- $X$  coefficient. The other matching has a nonzero all- $X$  coefficient, so adding the first cannot lower its displayed scale. At three, two one- $Y$  coefficient rows give a unit triangular minor at scale  $p^2$ . The Plucker relation has unit coefficients, including at two. Thus the named matching directions, at the displayed scales, are simultaneously bases of the integral Hodge lattice and of the ordinary-product lattice. Concretely, after the dyadic unramified split,

$$e_{0i} = 1, \quad e_{A_1 A_2} = e_{B_1 B_2} = 1, \quad e_{A_i B_j} = 2.$$

By Lemma 3.6, the only nontrivial squarefree width in dimension five is  $\ell = 2$ , precisely the four-slot calculation above. In codimension three, components of types  $(2, 2, 2)$  and  $(2, 2, 1, 1)$  are already product-saturated. A component of type  $(2, 1, 1, 1, 1)$  is a primitive diagonal factor times the four-slot calculation above. Codimensions four and five are obtained by adjoining primitive diagonal volume factors. Hence all  $Q_{J_b}^k$  vanish.  $\square$

The repeated roots are essential. The next statement gives a comparison in the same dimension where the divided-power conclusion of Theorem 3.3 holds but the ambient Hodge lattice is strictly larger.

**Proposition 3.8** (A sharp neighboring defect). *Let  $E$  be non-CM, and let a one-depth finite-étale marked graph presentation on  $E^5$  have depth  $p^a$  and five pairwise distinct residual roots after finite unramified splitting. Then*

$$Q_A^2 \simeq (\mathbf{Z}/p^a)^5, \quad Q_A^3 \simeq (\mathbf{Z}/p^a)^5, \quad Q_A^k = 0 \quad (k = 0, 1, 4, 5). \quad (3.14)$$

*Nevertheless every divided power of a divisor remains an ordinary product.*

*Proof.* On a four-subset  $J$ , choose the all- $X$ -cancelling direction  $h_J$  and a complementary matching direction  $r_J$ . A one- $Y$  coefficient with a unit root difference gives the exact lattices

$$I_J = O p^{3a} h_J \oplus O p^{4a} r_J, \quad P_J = O p^{4a} h_J \oplus O p^{4a} r_J.$$

Thus each support contributes exactly one copy of  $O/p^a$ , and the five four-subsets are distinct multidegree summands. Multiplication by the primitive volume class on the complementary slot gives the five independent codimension-three copies directly. The remaining multidegrees are saturated. The final assertion is Theorem 3.3; it shows why divided-power saturation and ambient integral Hodge saturation are different statements.  $\square$

Voisin identifies the primitive minimal class as the exact obstruction needed for cubic threefolds.

**Corollary 3.9** (Universal  $CH_0$ -triviality). *Every smooth member of the nonstandard  $A_5$ -cubic pencil is universally  $CH_0$ -trivial.*

*Proof.* By Theorem 3.5, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin's Corollary 4.4 [17] says that, for a smooth complex cubic threefold, this is equivalent to universal  $CH_0$ -triviality.  $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.9.

## 4 Framed formal monodromy

The obstruction used here belongs to the small even quantum connection itself. We follow it through each blowup in a weak factorization, without passing to equivalence classes of quantum atoms. This avoids a subtlety: a formal bulk gauge is coefficientwise Laurent in the loop coordinate, but need not specialize to a meromorphic gauge at a nearby analytic point.

Let  $T$  be a smooth projective variety. Write

$$\Lambda_T = \mathbf{C}[[\mathrm{NE}_{\mathrm{num}}^{\mathbf{Z}}(T)]]$$

for the completed monoid ring of the image of effective homology classes in the numerical curve lattice

$$N_1(T) = \mathrm{im}(H_2(T, \mathbf{Z}) \longrightarrow \mathrm{Hom}(\mathrm{NS}(T)/\mathrm{tors}, \mathbf{Z})).$$

Thus  $N_1(T)$  is torsion-free and is separated by integral divisors. Let  $\mathcal{K}_T$  be an algebraic closure of its fraction field. The loop coordinate  $z$  remains independent: in particular,  $\mathcal{K}_T$  contains no chosen root of  $z$ . Apply the Levelt–Turrittin construction to the small even quantum connection over  $\mathcal{K}_T((z))$ . Its formal monodromy is the action of one turn on the original  $z$ -disc. Thus it includes both regular-singular monodromy and the deck permutation of ramified exponential factors.

**Definition 4.1.** Let  $M_f(T)$  be this framed formal-monodromy operator, and put

$$\nu_6(T) = \text{mult}_{e^{\pi i/3}} M_f(T) + \text{mult}_{e^{-\pi i/3}} M_f(T).$$

The multiplicities are algebraic multiplicities in the characteristic polynomial.

The characteristic polynomial is unchanged by extending one algebraic closure of the numerical Novikov fraction field to a common overfield. Thus  $\nu_6(T)$  is independent of that choice.

The frame matters. Adjoining roots of Novikov monomials is harmless, but adjoining a root of  $z$  would replace the original loop and could erase the deck part of the operator.

We first record two formal facts used in the comparison theorems. They also fix the completion convention in the proof.

**Lemma 4.2** (Coefficientwise base change). *Let  $D$  be a formal differential module over  $k((z))$ , and let  $B$  be a complete separated  $I$ -adic  $k$ -algebra on which  $d/dz$  is zero. Scalar extension in the coefficient direction preserves the characteristic polynomial of framed formal monodromy. The same holds after a connection gauge whose entries use only integral powers of  $z$ .*

*Proof.* Choose a Levelt–Turrittin solution algebra  $R$  for  $D$  with differential constants  $k$ . At every finite level, the constants of  $R \otimes_k B/I^N$  are  $B/I^N$ : expansion in a  $k$ -basis reduces this to  $R^{d/dz} = k$ . A fundamental solution matrix then identifies the horizontal module with

$$\text{Sol}(D) \otimes_k B/I^N.$$

Passing to the inverse limit gives  $\text{Sol}(D) \widehat{\otimes}_k B$ , with monodromy  $M_f(D) \otimes 1$ . An integral- $z$  gauge is single-valued on the original disc and conjugates this action.  $\square$

**Lemma 4.3** (Formal base shift). *Suppose a pullback of the even quantum connection of  $T$  is defined over a complete filtered coefficient ring  $B$ , and its bulk coordinate has the form*

$$a_0 \mathbf{1} + a_2^\circ + \eta, \quad a_2^\circ \in H^2(T) \otimes F^0 B, \quad \eta \in F^1 B \widehat{\otimes} H^{\text{ev}}(T).$$

*Assume that  $a_0$  and  $a_2^\circ$  are constant in the positive filtration. Its framed formal-monodromy characteristic polynomial equals that of the small connection after the divisor substitution*

$$Q^d \longmapsto e^{\langle a_2^\circ, d \rangle} Q^d. \quad (4.1)$$

*Proof.* The string equation removes  $a_0 \mathbf{1}$  by a scalar exponential in  $z^{-1}$ ; this is single-valued and has trivial framed monodromy. The divisor equation gives (4.1). It remains to remove  $\eta$ , including its component in  $H^2(T)$ .

Modulo  $F^N B$ , the flat bulk equations

$$\partial_{\eta_i} G = -z^{-1}(\phi_i \star_\eta) G, \quad G|_{\eta=0} = 1,$$

have an invertible solution with bounded Laurent order in  $z$ . These solutions are compatible in  $N$ . Their inverse limit belongs to the pro-Laurent gauge group

$$\varprojlim_N \text{GL}(H^{\text{ev}}(T) \otimes (B/F^N B)((z))). \quad (4.1a)$$

This ring is deliberately larger than  $B((z))$ : the lower Laurent bound may decrease with  $N$ . Formal monodromy is defined on this pro-system, and at each finite quotient the gauge uses only integral powers of  $z$ , hence is single-valued on the original disc. Lemma 4.2 makes the characteristic polynomial constant at every quotient; compatibility makes the same polynomial the inverse-limit invariant. No Levelt–Turrittin decomposition over a larger analytic field, and no evaluation at a nonnilpotent analytic bulk point, is used.  $\square$

We now apply Iritani's quantum blowup theorem and the Iritani–Koto projective-bundle theorem. Both are stated over completed homological Novikov rings. Iritani's subsequent notes record explicitly that the blowup change of variables and decomposition preserve cohomological parity [10, Section 2].

**Lemma 4.4** (Numerical Novikov base change). *The quotient from the effective homology monoid to its image in  $N_1(T)$  extends continuously to completed monoid rings, and the quantum connection and the two comparison theorems base-change along it.*

*Proof.* Filter both monoids by intersection with an ample divisor. Effective homology classes below a fixed cutoff form a finite set, as follows from the finite-type Chow varieties of cycles of bounded degree. The quotient is therefore defined on every truncated monoid ring and passes to the inverse limit. The quantum products descend by finite sums of coefficients with the same numerical class. Pushforward of curves, pairing with  $c_1(N_{C/T})$ , and the derivations  $Q\partial_Q$  depend only on numerical classes. Since the comparison theorems provide both a map and its inverse, they remain inverse after this base change.  $\square$

**Proposition 4.5** (Framed operation formulas). *Let  $V \rightarrow T$  be a vector bundle of rank  $r \geq 2$ . Then*

$$\nu_6(\mathbf{P}_T(V)) = r\nu_6(T). \quad (4.2)$$

*Let  $\tilde{T} = \text{Bl}_C T$ , where  $C \subset T$  is smooth of codimension  $c \geq 2$ . In Iritani's comparison field one has*

$$\nu_6(\tilde{T}) = \nu_6(T) + \sum_{j=0}^{c-2} \nu_6(C; \chi_j), \quad (4.3)$$

*where  $\chi_j$  is the numerical Novikov specialization of the  $j$ -th center summand. The endpoint terms in (4.3) are the intrinsic small invariants; the center terms are computed after the displayed specializations.*

*Proof.* Iritani's Theorem 5.18 gives a formal coordinate isomorphism and a quantum- $D$ -module isomorphism

$$\text{QDM}(\tilde{T}) \cong \text{QDM}(T) \oplus \text{QDM}(C)^{\oplus(c-1)}$$

over a Laurent extension in the exceptional Novikov variable [9, Theorem 5.18]. The isomorphism and its inverse use only integral powers of  $z$ , and the mirror coordinates are independent of  $z$ . The comparison field adjoins  $q^{-1/s_c}$ , where  $s_c = c - 1$  for even  $c$  and  $s_c = 2(c - 1)$  for odd  $c$ ; this is only a coefficient extension. The reconstruction coordinate below is  $u = q^{-1/(c-1)}$ . The center base shift is controlled before passage to the final Laurent chart. Put  $u = q^{-1/(c-1)}$  and use the reconstruction coordinates of [9, Section 5.8.2]:

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = -(c-1)\lambda_j + h_{C,j} + O(u). \quad (4.4)$$

For the  $j$ -th target summand, let  $R_j$  be the image of  $Q_C^d \mapsto Q^{i_*d} u^{\rho_C \cdot d}$ , together with  $u$  and  $s_j$ , and set

$$J_j = (u, \{Q^{i_*d} u^{\rho_C \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

The center connection is defined over this image ring by [9, Remark 5.6]. Formula (4.4), together with (5.45), (5.47), and the initial asymptotics (5.27)–(5.30), shows that after the unit term  $-(c-1)\lambda_j$  and fixed divisor  $h_{C,j}$  are removed, its target bulk coordinate lies in  $J_j H^*(C)$ . The noninjective center monomial map merely replaces the source ring by its image.

First normalize the target connection over  $\mathbf{C}((u))$  by the single-valued string scalar for  $-(c-1)\lambda_j$  and by the fixed divisor shift. The normalized connection descends to  $B_j$ . Apply Lemma 4.3 modulo every  $J_j^N$ , take the compatible inverse limit, and only then extend scalars to Iritani's Laurent comparison field and apply  $\Psi$ . The fixed-divisor substitution is a  $z$ -constant coefficient automorphism, so it does not change  $\nu_6$ ; it is well defined on the image because  $h_{C,j}$  is a scalar multiple of  $\rho_C$ . Thus each center target summand has the framed monodromy of its specialized small connection. The ambient target uses  $\tau = \tau^\circ + t$ , with  $\tau^\circ = O(q^{-1})$ , and the same finite-quotient argument. Direct-sum additivity gives (4.3).

The endpoint map is faithful: on numerical curve lattices

$$\bar{\beta} \mapsto (\phi_* \bar{\beta}, -E \cdot \bar{\beta})$$

is injective because  $N^1(\tilde{T}) = \phi^* N^1(T) \oplus \mathbf{Z}[E]$ . Hence the Laurent extension preserves the intrinsic endpoint characteristic polynomials.

The projective-bundle proof is parallel, again in reconstruction coordinates. Its comparison field adjoins  $q^{-1/r'}$ , with  $r' = r$  when  $r-1$  is even and  $r' = 2r$  when  $r-1$  is odd; the reconstruction coordinate remains  $u = q^{-1/r}$ . The extra root is again a coefficient extension. Put  $u = q^{-1/r}$ . For its  $j$ -th target summand let  $R_j$  be the image of  $Q_T^d \mapsto Q^d u^{c_1(V) \cdot d}$ , together with  $u$  and  $s_j$ , and set

$$J_j = (u, \{Q^d u^{c_1(V) \cdot d} : d \neq 0\}, \text{components of } s_j) \subset R_j, \quad B_j = \varprojlim_N R_j / J_j^N.$$

Proposition 5.6 and Section 5.8 of Iritani–Koto give

$$\varsigma_j = \varsigma_j^\circ + s_j, \quad \varsigma_j^\circ = r\lambda_j - \frac{2\pi i j}{r} c_1(V) + O(u) [11, \text{Proposition 5.6 and Section 5.8}].$$

Normalize the Laurent unit and fixed divisor terms first. The normalized connection descends to  $B_j$ ; apply Lemma 4.3 in its finite quotients, and then extend scalars to the Laurent comparison field. Again the divisor substitution is a  $z$ -constant coefficient automorphism; it factors through the displayed  $u$ -exponent. This gives  $r$  copies of the intrinsic base module and proves (4.2). The base Novikov map is the embedding in [11, (1.1) and Remark 5.2]. The theorem assumes  $V^\vee$  globally generated. Tensoring  $V$  by a sufficiently negative line bundle achieves this without changing  $\mathbf{P}_T(V)$  [11, Remark 1.2]; the resulting divisor shift preserves framed monodromy by Lemma 4.3. All other Novikov extensions are in the coefficient direction. After the odd bulk variables vanish, pullback, pushforward, Fourier transformation, and multiplication by the characteristic-class factors all preserve cohomological parity. Hence both decompositions restrict to the even quantum connections used here. Formula (4.2) is tautological when  $r = 1$ .  $\square$

The center map in (4.3) is not necessarily injective [9, (5.15)]. Thus generic vanishing for  $C$  does not by itself imply that its specialized term vanishes. The next lemma supplies the missing implication.

**Definition 4.6.** A monomial map  $\chi : \Lambda_T \rightarrow A$  is *strictly Novikov-admissible* if  $A$  is a complete separated valued domain with domain associated graded, every monomial has nonzero image, and

$$v(\chi(Q^d)) = \ell(d), \tag{4.5}$$

where  $\ell$  is positive and proper on the effective numerical monoid.

We write  $\nu_6(T; \chi)$  for the framed small-even invariant after scalar extension along such a map.

The center maps in (4.3) are strictly admissible. Give the exceptional Laurent variable valuation zero and use an ample divisor  $H$  on  $T$ . Then

$$v_H(Q^{i \cdot d} q^{-\rho_C \cdot d / (c-1)}) = (H|_C) \cdot d.$$

The restriction  $H|_C$  is ample, which gives positivity and properness; the associated graded Laurent monoid ring is a domain.

**Lemma 4.7** (Divisor tagging). *Let  $\chi : \Lambda_T \rightarrow A$  be strictly Novikov-admissible. If the intrinsic small even quantum connection of  $T$  has no primitive-sixth framed monodromy, neither does its specialization along  $\chi$ .*

*Proof.* Choose integral divisors  $D_1, \dots, D_\rho$  whose pairings separate  $N_1(T)$ , and introduce formal variables  $\mathbf{t} = (t_1, \dots, t_\rho)$ . Define

$$\chi_{\mathbf{t}}(Q^d) = \chi(Q^d) \exp\left(\sum_i t_i (D_i \cdot d)\right). \quad (4.6)$$

This map from  $\Lambda_T$  to  $A[[\mathbf{t}]]$  is injective. To prove this, take a nonzero series. Properness in (4.5) makes its lowest-valuation support finite. Its initial image is a finite combination of distinct exponential characters. The finitely many exponent vectors are distinct integral vectors. Choose  $a = (a_i) \in \mathbf{Z}^\rho$  outside the finitely many integral hyperplanes on which two dot products agree, and substitute  $t_i = a_i s$ . The resulting exponents are distinct integers. Their derivatives at  $s = 0$  form a Vandermonde matrix with nonzero integer determinant, which remains invertible in the characteristic-zero fraction field. The characters are therefore linearly independent. The domain hypothesis on the associated graded prevents cancellation of the initial terms.

Let  $k_\chi = \overline{\text{Frac } A}$ . After embedding the two fraction fields in a common algebraic closure, injectivity makes the  $\chi_{\mathbf{t}}$ -module a scalar extension of the intrinsic module. On the other hand, the divisor equation identifies (4.6) with the degree-two formal bulk pullback of the  $\chi$ -module over  $B = k_\chi[[\mathbf{t}]]$ . Apply the finite-quotient argument in Lemma 4.3 with  $\eta = \sum_i t_i D_i$ : its integral- $z$  bulk gauge removes the positive divisor component. Hence vanishing over the intrinsic field implies vanishing after  $\chi$ .  $\square$

**Proposition 4.8** (Low-dimensional vanishing). *Let  $T$  be a point, a smooth projective curve, or a smooth projective surface. For every strictly Novikov-admissible specialization  $\chi$ ,*

$$\nu_6(T; \chi) = 0. \quad (4.7)$$

*Proof.* We first prove the intrinsic generic statement. If  $K_T$  is nef, the proof of [12, Claim 6.15] applies coefficient by coefficient over the small numerical Novikov ring. Set

$$\text{Gr} = \frac{\text{Deg} - \dim T}{2}, \quad g = \text{Gr} + \frac{1}{2}E,$$

where  $E$  is the identity in cohomological degrees of parity opposite to  $\dim T$  and is zero in the other degrees. The operator  $g$  has integral eigenvalues. Coefficientwise in  $Q^\beta$ , multiplication by  $c_1(T)$  sends

$$H^i(T) \longrightarrow H^{i+2-2c_1(T) \cdot \beta}(T).$$

If  $K_T$  is nef, then  $c_1(T) \cdot \beta \leq 0$ , so the Gr-weight rises by  $1 - c_1(T) \cdot \beta \geq 1$ . Conjugation by  $z^g$  therefore turns  $\nabla + \frac{1}{2}E dz/z$  into a regular-singular connection with nilpotent residue.

Undoing the half-parity correction permits only the monodromy eigenvalues 1 and  $-1$ , never a primitive sixth root.

This treats curves of positive genus and minimal surfaces with nef canonical class. A point is immediate. The remaining minimal varieties are  $\mathbf{P}^1$ ,  $\mathbf{P}^2$ , and ruled surfaces; (4.2) reduces them to a point or a curve. Every nonminimal surface is obtained from a minimal surface by point blowups, and (4.3) adds only point summands. Hence the intrinsic generic invariant vanishes in every case. Lemma 4.7 gives (4.7).  $\square$

**Theorem 4.9** (Birational invariance through dimension four). *If  $Y$  and  $Y'$  are birational smooth projective complex varieties of the same dimension at most four, then*

$$\nu_6(Y) = \nu_6(Y'). \quad (4.8)$$

Consequently, for smooth projective threefolds,

$$Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1 \implies \nu_6(Y) = \nu_6(Y').$$

*Proof.* Weak factorization decomposes a birational map between smooth varieties into blowups and blowdowns with smooth centers of codimension at least two [1]. In dimension at most four every nontrivial center has dimension at most two. Proposition 4.8 makes every center term in (4.3) zero, so the endpoint values agree at each step.

If  $Y, Y'$  are smooth threefolds and  $Y \times \mathbf{P}^1 \dashrightarrow Y' \times \mathbf{P}^1$ , apply the first assertion in dimension four and (4.2):

$$2\nu_6(Y) = \nu_6(Y \times \mathbf{P}^1) = \nu_6(Y' \times \mathbf{P}^1) = 2\nu_6(Y').$$

Cancelling two in  $\mathbf{Z}$  proves the stabilization assertion.  $\square$

**Proposition 4.10** (The cubic packet). *For every smooth cubic threefold  $X$ ,*

$$\nu_6(X) = 2. \quad (4.9)$$

*Proof.* In the basis  $1, P, P^2, P^3$ , Cai explicitly block-diagonalizes the small even quantum connection. The rank-two zero-exponential block has indicial polynomial

$$\rho^2 + \rho + \frac{5}{36},$$

with roots  $-1/6$  and  $-5/6$  [3, Section 3 and Proposition 6]. His gauge uses only integral powers of  $z$ . The two corresponding framed-monodromy eigenvalues are  $e^{-\pi i/3}$  and  $e^{\pi i/3}$ . The two remaining rank-one even blocks have formal-monodromy factor 1, hence contribute no primitive sixth root. Thus the multiplicity is exactly two. The numerical curve lattice of a cubic threefold has rank one, so the numerical-Novikov convention does not change the calculation.  $\square$

*Proof of Theorem 1.2.* The projective-bundle formula and Proposition 4.10 give

$$\nu_6(X \times \mathbf{P}^1) = 2\nu_6(X) > 0. \quad (4.10)$$

Suppose that  $X \times \mathbf{P}^1$  were rational. By Theorem 4.9,

$$\nu_6(X \times \mathbf{P}^1) = \nu_6(\mathbf{P}^4) = 0,$$

where the last equality follows from (4.2) over a point. This contradicts (4.10).  $\square$



## 4.1 A genus-eight Fano consequence

Kuznetsov associates to every smooth prime Fano threefold  $V$  of genus eight a smooth Pfaffian cubic threefold  $X$ , together with rank-two vector bundles  $U$  on  $V$  and  $E^*$  on  $X$ , whose projectivizations are related by a flop [13, Theorems 2.17–2.18].

**Corollary 4.11** (One stabilization of  $V_{14}$ ). *For every smooth prime Fano threefold  $V$  of genus eight,*

$$\nu_6(V) = 2, \quad V \times \mathbf{P}^1 \text{ is irrational.} \quad (4.11)$$

*Proof.* The flop gives a birational map between the smooth projective fourfolds  $\mathbf{P}_V(U)$  and  $\mathbf{P}_X(E^*)$ . By Theorem 4.9, Proposition 4.5, and Proposition 4.10,

$$2\nu_6(V) = \nu_6(\mathbf{P}_V(U)) = \nu_6(\mathbf{P}_X(E^*)) = 2\nu_6(X) = 4.$$

Hence  $\nu_6(V) = 2$ . Formula (4.2) now gives  $\nu_6(V \times \mathbf{P}^1) = 4$ , whereas every rational smooth fourfold has  $\nu_6 = 0$  by Theorem 4.9.  $\square$

The argument is sharp for this invariant. In a factorization of the fivefold  $X \times \mathbf{P}^2$ , a center may have dimension three and may itself carry the cubic packet. The next stabilization therefore requires information about the packet’s Stokes or Rees placement, not only its multiplicity.

## 5 The separation theorem

*Proof of Theorem 1.1.* Corollary 3.9 gives universal  $CH_0$ -triviality of  $X_b$ . The projective bundle formula for Chow groups gives the same property for  $X_b \times \mathbf{P}^1$ . Theorem 1.2 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the  $A_5$ -component.  $\square$

The theorem separates two tests on one variety. The algebraic class  $\Theta^4/4!$  removes the decomposition-of-diagonal obstruction. The primitive-sixth framed-monodromy multiplicity survives one stabilization and obstructs a rational factorization. The first application is special to the exotic  $A_5$ -family, although the all-degree graph-saturation theorem applies to marked finite-etale elliptic-power quotients more generally. The second mechanism is a birational invariant through dimension four; besides every cubic threefold, it gives one-step irrationality for every smooth genus-eight prime Fano threefold. The  $A_5$ -family is where the cycle and quantum conclusions hold simultaneously.

The result stops after one stabilization. For  $X \times \mathbf{P}^m$  with  $m \geq 2$ , weak-factorization centers may have dimension at least three and can carry primitive-sixth quantum packets. The low-dimensional vanishing in Proposition 4.8 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

Thus the same smooth fourfolds pass the universal  $CH_0$ -triviality test and fail the framed-quantum rationality test. Vanishing of the decomposition-of-the-diagonal obstruction does not predict rationality even after one explicit stabilization.

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